Python Programming - 2301CS404

Lab - 6

Roll No.: 514

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01) WAP to find sum of all the elements in a List.

In [1]:
```
num =int(input("Enter The element Number of List: "))
li = list()
for i in range(1, num+1,1):
    ele = int(input(f"Enter The {i} element of List: "))
    li.append(ele)
print(li)
print("Sum:",sum(li))
```

```
[1, 2, 3, 4, 5]
Sum: 15
```

02) WAP to find largest element in a List.

In [19]:
```
li = [10,20,30,40,50]

large = li[0]
for i in range(1 , len(li),1):
    if(large < li[i]):
        large = li[i]
print("Largest Number:",large)
```

```
Largest Number: 50
```

03) WAP to find the length of a List.

In [3]:
```
l2 = [100,200,300,405,65,8,6,5]
l2Length = len(l2)
print("Length Of list:",l2Length)
```

```
Length Of list: 8
```

04) WAP to interchange first and last elements in a list.

In [14]:
```
l3 = [100,200,300,400,500]
l3.sort(reverse=True)
print(l3)
```

```
[500, 400, 300, 200, 100]
```

05) WAP to split the List into two parts and append the first part to the end.

In [41]:
```
l4 = [10,20,30,40,50,60]
#l4[40,50,60,10,20,30]
lefthalf = l4[0:4]
righthalf = l4[4:6]
print(lefthalf)
print(righthalf)

newl4 = list()
newl4.append(righthalf)
newl4.append(lefthalf)
print("new List: ",newl4)
```

```
[10, 20, 30, 40]
[50, 60]
new List:  [[50, 60], [10, 20, 30, 40]]
```

06) WAP to interchange the elements on two positions entered by a user.

In [33]:
```
li6 = [10,20,30,40,50,60,70]
print("List:",li6)
pos1 = int(input("Enter The element position of change element: "))
pos2 = int(input("Enter The element position of change element: "))
temp = 0
temp = li6[pos1]
li6[pos1] = li6[pos2]
li6[pos2] = temp
print(li6)
```

```
List: [10, 20, 30, 40, 50, 60, 70]
[10, 50, 30, 40, 20, 60, 70]
```

07) WAP to reverse the list entered by user.

In [28]:
```
num2 =int(input("Enter The element Number of List: "))
li7 = list()
for i in range(1, num2+1,1):
    element = int(input(f"Enter The {i} element of List: "))
    li7.append(element)
print(li7)
li7reversed = li7.reverse()
print("Reversed List:",li7reversed)
```

```
[6, 4, 8, 2, 1]
Reversed List: None
```

08) WAP to print even numbers in a list.

In [65]:
```
li8 = []
start = int(input("Enter The Starting Value: "))
end = int(input("Enter The ending Value: "))
for i in range(start,end+1,1):
    if(start % 2 == 0):
        li8.append(start)
    start = start + 1
print(li8)
```

```
[2, 4, 6, 8, 10, 12, 14]
```

09) WAP to count unique items in a list.

In [70]:
```
li9 = [1, 2, 2, 5, 8, 4, 4, 8]
newli9 = []

count = 0

for i in li9:
    if i not in newli9:
        count += 1
        newli9.append(i)

print("No of unique items are:", count)
```

```
No of unique items are: 5
```

10) WAP to copy a list.

In [48]:
```
li10 = [10,2,4,5,6,8,54]
li10Copy = li10.copy()
print("List:",li10)
print("Copy:",li10Copy)
```

```
List: [10, 2, 4, 5, 6, 8, 54]
Copy: [10, 2, 4, 5, 6, 8, 54]
```

11) WAP to print all odd numbers in a given range.

In [66]:
```
li11 = []
starting = int(input("Enter The Starting Value: "))
ending = int(input("Enter The ending Value: "))
for i in range(starting,ending+1,1):
    if(starting % 2 == 1):
        li11.append(starting)
    starting = starting + 1
print(li11)
```

```
[1, 3, 5, 7, 9, 11, 13, 15]
```

12) WAP to count occurrences of an element in a list.

In [3]:
```
li12 = [1, 2, 3, 4, 1, 2, 1, 5, 6, 1]
elementCount = 1
```

```
count2 = li12.count(elementCount)
print(f"The element {elementCount} occurrences {count2} times.")
```

```
The element 1 occurrences 4 times.
```

13) WAP to find second largest number in a list.

In [7]:
```
a = [10, 20, 4, 65, 99]
Large = secondLarge = float()
for n in a:
    if n > Large:
        secondLarge = Large
        Large = n
    elif n > max2 and n != Large:
        secondLarge = n
print(secondLarge)
```

```
65
```

14) WAP to extract elements with frequency greater than K.

In [3]:
```
T14 = (1, 2, 2, 3, 3, 3, 4, 4)
K = 2

newDict = {}
for x in T14:
    newDict[x] = newDict.get(x, 0) + 1

output = tuple(x for x, c in newDict.items() if c >= K)

print(output)
```

```
(2, 3, 4)
```

15) WAP to create a list of squared numbers from 0 to 9 with and without using List Comprehension.

In [57]:
```
li15 = [0,1,2,3,4,5,6,7,8,9]
newli15 = list()
for i in range(0,len(li15),1):
    newli15.append(i**2)
print("Using for loop: ",newli15)
[i**2 for i in range(0,10,1)]
```

```
Using for loop:  [0, 1, 4, 9, 16, 25, 36, 49, 64, 81]
```

Out[57]:
```
[0, 1, 4, 9, 16, 25, 36, 49, 64, 81]
```

16) WAP to create a new list (fruit whose name starts with 'b') from the list of fruits given by user.

In [5]:
```
fruits = input("Enter fruits separated by space: ").split()

li16 = [i for i in fruits if i.lower().startswith('b')]

print("Fruits starting with 'b':", li16)
```

```
Fruits starting with 'b': ['Banana']
```

17) WAP to create a list of common elements from given two lists.

In [8]:

```
li17 = [10,20,30,40,50]
li18 = [30,40,60,70]
new_li = []
for i in li17:
    if i in li18:
        new_li.append(i)
print(new_li)
```

```
[30, 40]
```

In [ ]: